AOS-CX 10.06 Quality of Service Guide

8400 Switch Series

a Hewlett Packard
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Notices

Acknowledgments

Chapter 1 About this document

This document describes features of the AOS-CX network operating system. It is intended for administrators responsible for installing, configuring, and managing Aruba switches on a network.

Applicable products

This document applies to the following products:

Aruba 8400 Switch Series (JL375A, JL376A)

Latest version available online

Updates to this document can occur after initial publication. For the latest versions of product documentation, see the links provided in **Support and other resources**.

Command syntax notation conventions

Convention	Usage
`example-text`	Identifies commands and their options and operands, code examples, filenames, pathnames, and output displayed in a command window. Items that appear like the example text in the previous column are to be entered exactly as shown and are required unless enclosed in brackets (`[ ]`).
`example-text`	In code and screen examples, indicates text entered by a user.
Any of the following: • *<example-text>* • <example-text> • *example-text* • *example-text*	Identifies a placeholder—such as a parameter or a variable—that you must substitute with an actual value in a command or in code: • For output formats where italic text cannot be displayed, variables are enclosed in angle brackets (< >). Substitute the text—including the enclosing angle brackets—with an actual value. • For output formats where italic text can be displayed, variables might or might not be enclosed in angle brackets. Substitute the text including the enclosing angle brackets, if any, with an actual value.
`|`	Vertical bar. A logical `OR` that separates multiple items from which you can choose only one. Any spaces that are on either side of the vertical bar are included for readability and are not a required part of the command syntax.
`{ }`	Braces. Indicates that at least one of the enclosed items is required.

Table Continued

Convention	Usage
`[ ]`	Brackets. Indicates that the enclosed item or items are optional.
… or `. . .`	Ellipsis: • In code and screen examples, a vertical or horizontal ellipsis indicates an omission of information. • In syntax using brackets and braces, an ellipsis indicates items that can be repeated. When an item followed by ellipses is enclosed in brackets, zero or more items can be specified.

About the examples

Examples in this document are representative and might not match your particular switch or environment.

The slot and port numbers in this document are for illustration only and might be unavailable on your switch.

Understanding the CLI prompts

When illustrating the prompts in the command line interface (CLI), this document uses the generic term `switch`, instead of the host name of the switch. For example:

```
switch>
```

The CLI prompt indicates the current command context. For example:

`switch>`

Indicates the operator command context.

`switch#`

Indicates the manager command context.

`switch(`*`CONTEXT-NAME`*`)#`

Indicates the configuration context for a feature. For example:

```
switch(config-if)#
```

Identifies the `interface` context.

Variable information in CLI prompts

In certain configuration contexts, the prompt may include variable information. For example, when in the VLAN configuration context, a VLAN number appears in the prompt:

```
switch(config-vlan-100)#
```

When referring to this context, this document uses the syntax:

```
switch(config-vlan-<VLAN-ID>)#
```

Where *`<VLAN-ID>`* is a variable representing the VLAN number.

Identifying switch ports and interfaces

Physical ports on the switch and their corresponding logical software interfaces are identified using the format:

`member/slot/port`

On the 8400 Switch Series

- *member*: Always 1. VSF is not supported on this switch.
- *slot*: Specifies physical location of a module in the switch chassis.
  - Management modules are on the front of the switch in slots 1/5 and 1/6.
  - Line modules are on the front of the switch in slots 1/1 through 1/4, and 1/7 through 1/10.
- *port*: Physical number of a port on a line module

For example, the logical interface `1/1/4` in software is associated with physical port 4 in slot 1 on member 1.

Identifying modular switch components

- Power supplies are on the front of the switch behind the bezel above the management modules. Power supplies are labeled in software in the format: *member/power supply*:
  - *member*: 1.
  - *power supply*: 1 to 4.
- Fans are on the rear of the switch and are labeled in software as: *member/tray/fan*:
  - *member*: 1.
  - *tray*: 1 to 4.
  - *fan*: 1 to 4.
- Fabric modules are not labeled on the switch but are labeled in software in the format: *member/module*:
  - *member*: 1.
  - *member*: 1 or 2.
- The display module on the rear of the switch is not labeled with a member or slot number.

Quality of Service (QoS) enables network administrators to customize how different types of traffic are serviced on a network, taking into account the unique characteristics of each traffic type and its importance within an organization's infrastructure. QoS ensures uniform and efficient traffic handling, keeping the most important traffic moving at an acceptable speed, regardless of current bandwidth usage. It also provides methods for administrators to control the priority settings of inbound traffic arriving at each network device.

End-to-end QoS behavior

The QoS settings on each network device must be aligned to achieve the desired end-to-end QoS behavior for a network. Three service types can be used to categorize and prioritize network traffic:

- Best Effort Service
- Ethernet Class of Service (CoS)
- Internet Differentiated Services (DiffServ)

For a network as a whole, it is best to select one service type to use as the primary end-to-end behavior, and then use the other two service types as needed.

Best effort service

This is the simplest service type. All traffic is treated equally in a first-come, first-served manner. If the traffic load is low in relation to the capacity of the network links, then there is no need for the administrative complexity and costs of maintaining a more complex end-to-end policy. This is sometimes called over-provisioning, as all link speeds are much higher than peak loads on the network.

Class of Service

Class of Service (CoS) is a method for classifying network traffic at layer 2 by marking 802.1Q VLAN Ethernet frames with one of eight service classes.

CoS	Traffic type	Example protocols
7	Network Control	STP, PVST
6	Internetwork Control	BGP, OSPF, PIM
5	Voice (<10ms latency)	VoIP(UDP)
4	Video (<100ms latency)	RTP
3	Critical Applications	SQL RPC, SNMP
2	Excellent Effort	NFS, SMB
0	Best Effort	HTTP, TELNET
1	Background	SMTP, IMAP

CoS 1 is deliberately set as the lowest CoS. This enables a traffic service level below the default (best effort) traffic level to be specified.

The 3-bit Priority Code Point (PCP) field within the 16-bit Ethernet VLAN tag is used to mark the CoS.

```
+--------+--------+--------+----------+-----------+--------
| mac-da | mac-sa | 0x8100 | VLAN tag | ethertype | data...
+--------+--------+--------+----------+-----------+--------
                          /            \
                         /              \
                        /                \
                     +-----+-----+---------+
                     | pcp | dei | vlan_id |
                     +-----+-----+---------+
```

Differentiated services

Differentiated services (DiffServ) is a method for classifying network traffic at layer 3 by marking packets with one of 64 different service classes. Services classes are identified by the Differentiated services Code Point (DSCP) value. Some common DSCP values are:

DSCP	Name	Service class	RFC
56	CS6	Network Control	2474
46	EF	Telephony	3246
40	CS5	Signaling	2474
34, 36, 38	AF41, AF42, AF43	Multimedia Conferencing	2597
32	CS4	Real-Time Interactive	2474
26, 28, 30	AF31, AF32, AF33	Multimedia Streaming	2597
24	CS3	Broadcast Video	2474
18, 20, 22	AF21, AF22, AF23	Low-Latency Data	2597
16	CS2	OAM	2474
00	CS0,BE,DF	Best Effort	2474
10, 12, 14	AF11, AF12, AF13	Bulk Data	2597
08	CS1	Low-Priority Data	3662

DSCP CS1 (08) CoS 1 is deliberately set as the lowest priority. This enables a traffic service level below the standard (best effort or default forwarding) level to be specified.

The DSCP value is carried within the IPv4 DSCP field or the upper 6-bits of the 8-bit IPv6 Traffic Class (TC) field.

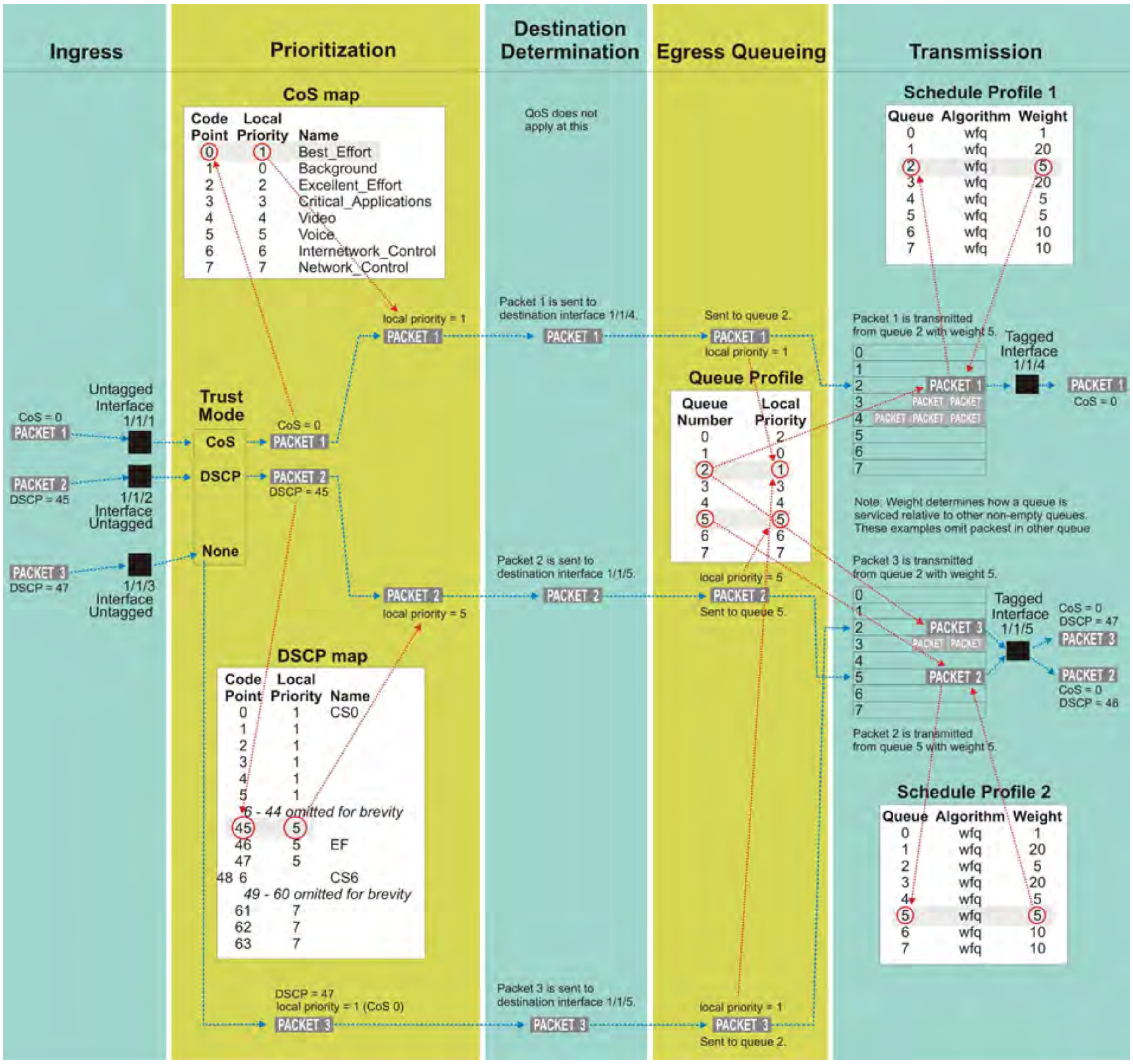
```
     +-----+------+-----+-----+----+--------+-----+-------+--------+-------+-------+--------
IPv4 | ver | dscp | ecn | len | id | offset | ttl | proto | chksum | ip-sa | ip-da | data...
     +-----+------+-----+-----+----+--------+-----+-------+--------+-------+-------+--------


         +------+-----+
         | dscp | ecn |
         +------+-----+
          \         /
           \       /
      +-----+-----+-----+-------+-------------+-----------+-------+-------+--------
 IPv6 | ver | tc  | len | label | next_header | hop_limit | ip-sa | ip-da | data...
      +-----+-----+-----+-------+-------------+-----------+-------+-------+--------
```

QoS on the switch

There are five key stages a packet passes through when traversing a switch: ingress, prioritization, destination determination, egress queuing, and transmission. The following table provides an overview of each stage, and lists the commands that can be used to configure QoS settings.

Ingress	Prioritization	Destination Determination	Queueing	Transmission
Packets arrive at a switch interface (port).	A local priority value is assigned to the packet. This is used internally by the switch to classify traffic.	The egress interface for each packet is determined to allow queuing and packets are sent to the appropriate interface. QoS does not affect this stage.	Packets are queued based on the selected destination interface and local priority to queue mapping.	A scheduler defines the order in which packets are selected from the queues and are transmitted on the egress interface.
rate-limit Rate limiting can control ingress traffic flow by packet type.	**qos trust** Determines which packet values are used to create local priority: CoS, DSCP, or none (in which case the default value of CoS 0 is used). **qos cos-map** Defines how CoS values are mapped to local priority values. **qos dscp-map** Defines how DSCP values are mapped to local priority values.		**qos queue-profile** Creates a queue profile which defines eight transmit queues. **map queue** Assigns a local priority value to a queue number. Packets with a matching local priority value are placed in the queue. **name queue** Assigns a name to a queue. **apply qos** Assigns a queue profile and schedule profile globally to all interfaces, or a schedule profile to a specific interface.	**qos schedule-profile** Sets the order that queues are selected to transmit packets. **qos threshold-profile** Applies ECN to queue packets that exceed a threshold. **qos shape** Limits egress bandwidth on an interface. **strict queue** Assigns the strict priority algorithm to a queue number in a schedule profile **wfq queue** Assigns the WFQ algorithm to a queue number in schedule profile.

The following diagram shows how different packets might traverse a switch. It also shows how QoS configuration settings apply at each stage.

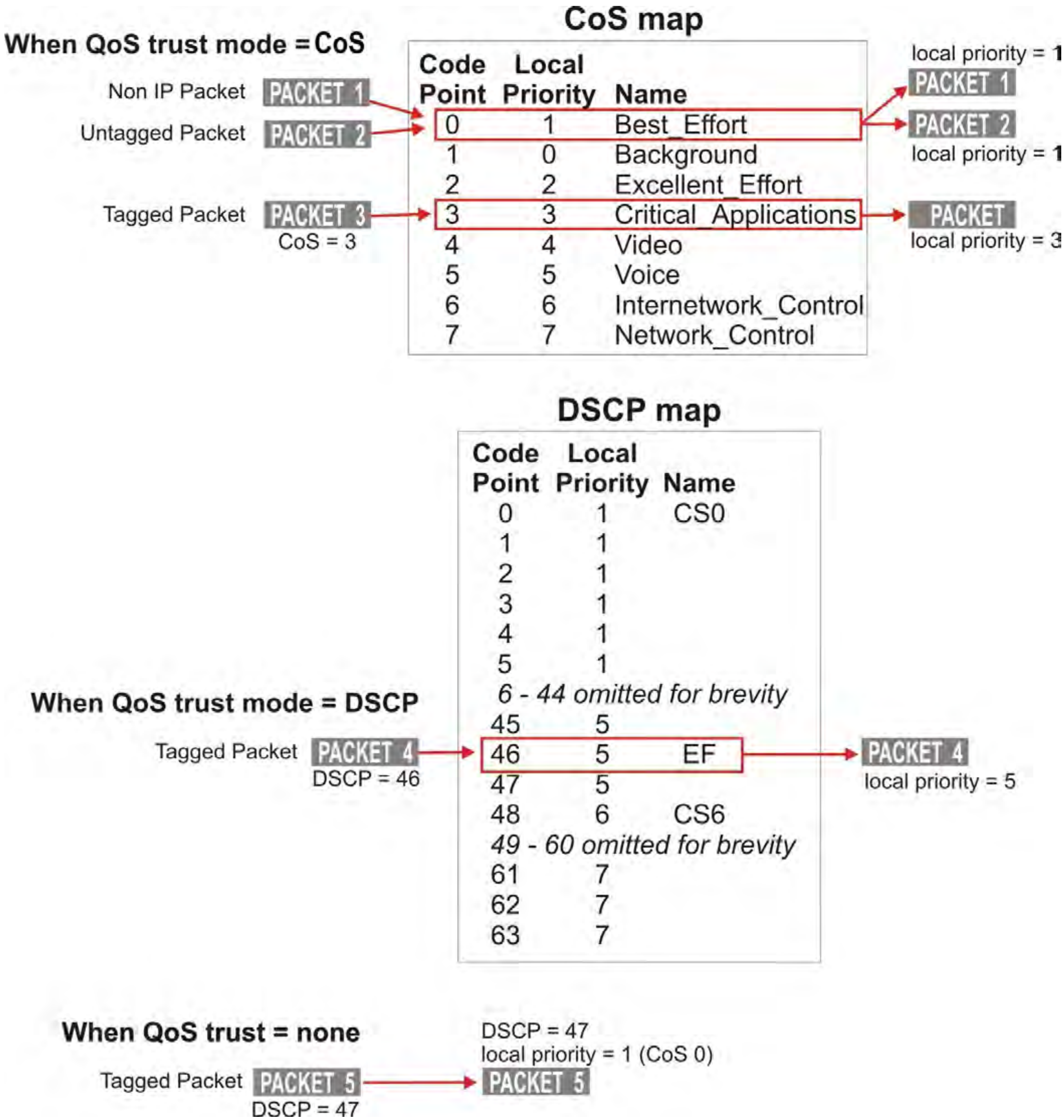


QoS trust

Traffic priorities for networks can be carried in VLAN tags, using the CoS Priority Code Point (PCP), or in IP packet headers, using the Differentiated Services Code Point (DSCP). Whether these priorities affect how traffic is serviced, depends on how QoS trust mode is configured on the switch. QoS trust mode specifies how the switch assigns local priority values to ingress packets. Trust mode can be set globally for all interfaces, or individually for each interface. By default, trust mode is set to `none`, meaning that any QoS information in the packet (CoS or DSCP) is ignored, and local priority values are assigned from the CoS map value for code point 0.

When trust mode is set to CoS or DSCP, the switch translates the QoS settings in VLAN tags (for CoS), or the DS field in an IP header (for DSCP), to local priority values on the switch. Translation is controlled by the CoS map or DSCP map tables.

For example:



Dynamic QoS trust mode

The device profile feature can dynamically set the QoS trust mode on an interface based on the LLDP information exchanged with a link partner. The device profile's trust mode temporarily overrides the static trust mode configured for an interface. The override remains in place as long as that link partner is connected and its link state is **up**. Use command `show interface IFNAME qos` to view the current QoS trust mode for an interface.

Port rate limiting

Port rate limiting helps control undesirable traffic. Its purpose is to allow enough broadcast, multicast, and ICMP rate-limit traffic for the network to function properly, while preventing flooding and traffic storms.

A certain amount of each type of traffic is required for normal network operation. Broadcast packets may include ARP and DHCP traffic, for instance. Video streams, and certain types of network protocol packets, are multicasts. Unknown-unicast packets may be intended for devices whose addresses have temporarily aged out of network-forwarding caches. Configuring rate limits can help provide the balance between necessary and flooded traffic.

Queue profile

A queue profile defines the queues that are associated with an interface to control the transmission of packets. Each profile supports up to eight queues, numbered 0 to 7. The larger the queue number, the higher its priority during transmission scheduling. Packets are assigned to a queue based on their local priority value (0 to 7). A queue profile must map all eight local priority values to whatever queues are being used on the switch, and a schedule profile must specify the configuration for those same queues. A queue without a local priority value assigned to it is not used to store packets.

The switch is automatically provisioned with an initial queue profile named `factory-default` which assigns each local priority to the queue of the same number. To see the default queue profile, use the command `show qos queue-profile factory-default`:

```
switch# show qos queue-profile factory-default
queue_num local_priorities name
--------- ---------------- ----
0         0                Scavenger_and_backup_data
1         1
2         2
3         3
4         4
5         5
6         6
7         7
```

More than one local priority value can be assigned to the same queue. For example,

Local Priority	Queue
0	0
1	1
2	2
3	3
4	4
5	5
6	5
7	5

Commonly used commands for working with QoS queues are as follows:

- `qos queue-profile`: Creates an empty queue-profile and enters the profile configuration context.
- `name queue`: Assigns a descriptive name to a queue.
- `map queue`: Assigns a local-priority to a queue.
- `apply qos queue-profile`: Applies a queue-profile globally to all interfaces.

Schedule profiles

A schedule profile determines the order in which queues are selected for transmission, and the amount of service available for each queue. A schedule profile must be configured on every interface at all times. A schedule profile can be applied globally to all interfaces, or only to specific interfaces.

Three options are available:

- All queues use weighted fair queuing (WFQ)
- All queues use strict priority queuing
- The highest priority queue uses strict priority, and all other queues use WFQ

A weighted schedule profile assigns relative servicing for each queue. The amount of service per weight is relative to the underlying hardware implementation, and to the weights assigned to the other non-empty queues. Strict scheduling can be used to service queues purely on the basis of highest priority first (at the risk of starving lower-priority queues during high stress periods). A combination of strict and weighted scheduling offers more service to the highest priority queue when needed, while preserving scheduling between the remaining queues, thus decreasing the risk of starvation.

The switch is automatically provisioned with a schedule profile named **`factory-default`**, which assigns DWRR to all queues with a weight of 1. Use the command `show schedule-profile factory-default` to view the default schedule profile. (Do not use `show running-configuration`, as it only displays changes from the initial settings.)

```
switch# show qos schedule-profile default
queue_num algorithm weight
--------- --------- ------
0         wfq       1
1         wfq       1
2         wfq       1
3         wfq       1
4         wfq       1
5         wfq       1
6         wfq       1
7         wfq       1
```

Egress queue shaping

Egress queue shaping limits the amount of traffic transmitted per strict output queue. The buffer associated with each egress queue stores excess traffic to absorb bursts and smooths the output rate. For example, an administrator might limit strict-priority queue traffic to prevent low-priority queue starvation in the event that a device inappropriately sends too many higher-priority packets.

Egress queue shaping can be configured on an Ethernet port or on a link aggregation group (LAG). To configure egress queue shaping, define a schedule profile with the strict priority algorithm assigned to each queue.

Egress port shaping

Egress port shaping limits the amount of aggregate traffic transmitted through a port.

To be effective, the egress port-shaping rate must be less than the port's line rate. By default, the egress port-shaping rate is the same as the line-rate of the port. Buffers associated with each port store excess traffic.

When both egress port-shaping and egress queue-shaping are configured on the same interface, the switch respects the minimum of both configurations.

Explicit Congestion Notification

Explicit Congestion Notification (ECN) provides a mechanism for two end-points to exchange end-to-end notification of network congestion. ECN uses a 2-bit field in the IP header to indicate that the traffic load on network equipment in the path between an ECN-capable sender and receiver is causing packets to be buffered, as defined by IETF RFC 3168 (https://tools.ietf.org/html/rfc3168).

Threshold profiles

Threshold profiles configure individual queue utilization thresholds as triggers for taking action (i.e., ECN marking) on a packet. A threshold profile is applied per-port and defines the threshold and action for each queue. Omitting configuration for a queue in a threshold profile means that queue will not be configured with a threshold value or action.

In an environment where responsive transport protocols are in use and congestion management features are required to reduce latency, ECN can be configured on queues carrying delay-sensitive traffic. The result is that queue utilization is actively managed, resulting in ECT packets being CE marked when queue utilization reaches or exceeds a configured threshold.

Virtual output queues

The switch uses a virtual output queue (VOQ) architecture where most packet buffering occurs on the ingress line module. Traffic destined for one port (unicast), uses different buffering and scheduling than traffic destined for multiple ports (flood). The relative priority and the amount of packet transmission for these two types of traffic differ.

For unicast traffic, each line module contains eight VOQs for every destination port in the chassis (one per local priority). The queue profile determines which VOQ is used to buffer a local priority. The schedule profile determines the order of VOQ servicing. Unicast packets wait in VOQs until the scheduler selects them to cross the fabric. Each destination port has a shallow egress transmit queue that buffers unicast packets.

Broadcast, multicast, and unknown-unicast packets, collectively called flooded traffic, use a separate path to the destination ports. Each line module contains eight VOQs per destination line module (including itself) to buffer traffic to be flooded (one VOQ per local priority). A copy of the packet to be flooded is buffered in VOQs for each destination line module. The queue profile determines which VOQ is used for each local priority.

Flooded traffic VOQs use a fixed strict schedule profile to determine the order of VOQ servicing. Flooded packets wait in VOQs until the scheduler selects them to cross the fabric. On the destination line module, they are replicated to one or more destination ports. Each destination port has a second shallow egress queue for replicated packets buffered for transmit.

A WFQ scheduler is used to select packets for transmission from the unicast and replicated traffic egress queues. When selecting packets between two non-empty queues, WFQ uses a fixed data weight of four for unicast traffic, and a weight of one for replicated traffic. As long as both queues are non-empty, replicated (flooded) packets comprise approximately 20% of the transmitted traffic, independent of the unicast scheduled percentages.

Terms

Class

For networking, a set of packets sharing a common characteristic. For example, all IPv4 packets.

Code point

The name of a packet header field, or the value carried within a packet header field:

Example 1: Priority code point (PCP) is the name of a field in the IEEE 802.1Q VLAN tag.

Example 2: Differentiated services code point (DSCP) is the name of a field carried within the DS field of an IP packet header.

Color

A metadata label associated with each packet within the switch. It has three values: green (0), yellow (1), or red (2). When packets encounter congestion for a resource (queue), the switch uses packet color to distinguish which packets must be dropped, and is mostly used for packets marked with Assured Forwarding (AF) DSCP values.

Not supported in this release.

Class of service (CoS)

A 3-bit value used to mark packets with one of eight classes (levels of priority). It is carried within the priority code point (PCP) field of the IEEE 802.1Q VLAN tag.

Differentiated services code point (DSCP)

A 6-bit value used to mark packets for different per-hop behavior as originally defined by IETF RFC 2474. It is carried within the differentiated services (DS) field of the IPv4 or IPv6 header.

Local priority

A meta-data label associated with a packet within the switch which is used to classify packets for different treatment (such as queue assignment). Eight local priorities are defined on the switch, numbered from 0 to 7. A queue profile must map all eight local priorities to whatever queues are in use on the switch, and a schedule profile must specify the configuration for these same queues.

Metadata

Information labels associated with each packet in the switch, separate from the packet headers and data. These labels are used by the switch in its handling of the packet. For example: arrival port, egress port, VLAN membership, and local priority.

Priority code point (PCP)

The name of a 3-bit field in the IEEE 802.1Q VLAN tag. It carries the CoS value to mark a packet with one of eight classes (priority levels).

Quality of service (QoS)

General term used when describing or measuring performance. For networking, it means how different classes of packets are treated when traversing a network or device.

Traffic class (TC)

General term for a set of packets sharing a common characteristic. It used to be the name of an 8-bit field in the IPv6 header originally defined by IETF RFC 2460. This field name was changed to differentiated services by IETF RFC 2474.

Type of service (ToS)

General term when there are different levels of treatment (fare class). It used to be the name of an 8-bit field in the IPv4 header originally defined by IETF RFC 791. This field name was changed to differentiated services by IETF RFC 2474.

Configuring QoS

Procedure

1. Configure how local priority values are assigned to ingress packets with the commands **qos cos-map** , **qos dscp-map**, and **qos trust**.
2. Optionally, add a rate limit for ingress traffic on one or more interfaces with the command **rate-limit**.
3. If you do not want to use the default QoS queue profile to map local priority to queue, create one or more custom queue profiles with the command **qos queue-profile**. For each queue in a custom queue profile:
   a. Assign a local priority value with the command **map queue**.
   b. Optionally, define a descriptive name with the command **name queue**. All local priorities (0 to 7) must be mapped to a queue, and the queues selected for use must be in contiguous order starting at 0.
4. If you do not want to use the default QoS schedule profile to determine the order in which queues are selected to transmit a packet, create one or more custom schedule profiles with the command **qos schedule-profile**. For each queue in a custom schedule queue profile, define scheduling priority with the commands **strict queue** and **wfq queue**.
5. Optionally, create a threshold profile to limit throughput on one or more queues with the command . Assign threshold values to the queue with the command . Then, apply the profile with the command **apply qoes threshold-profile**
6. Optionally for strict queues, configure egress queue shaping to limit egress bandwidth on an interface to a value that is less than its line rate. Use the `max-bandwidth` parameter of the **strict queue** command.
7. Activate QoS settings with the command **apply qos**. This command lets you apply a queue profile and schedule profile globally to all interfaces, or a schedule profile override to individual interfaces.

   When applying QoS settings to a port configured to support priority-based flow control, specific configuration settings must be respected when defining a CoS map and queue profile. See the command `flow-control` in the *ArubaOS-CX Command-Line Interface Guide* for details.
8. View QoS configuration settings with the provided `show` commands.

Examples

This example creates the following configuration:

- Configures CoS to be used to assign local priority to ingress packets.
- Modifies the default CoS map to assign CoS 1 to local priority **1**.
- Creates a queue profile named `Q1` and assigns local priorities as follows:

Queue	Local Priority
0	0
1	1
1	2
3	3
4	4
5	5
5	6
5	7

- Creates a schedule profile named `S1` and assigns WFQ to all queues in the schedule profile with the following weights:

Queue	Weight
0	5
1	10
3	15
4	25
5	50

- Creates a threshold profile named `T1` with the following limits:

Queue	Threshold
4	40 KB
5	50 KB

Applies **Q1** and **S1** to all interfaces that do not have a QoS override applied.

```
switch(config)# qos trust cos
switch(config)# qos cos-map 1 local-priority 1
switch(config)# qos queue-profile Q1
switch(config)# map queue 0 local-priority 0
switch(config)# map queue 1 local-priority 1
switch(config)# map queue 1 local-priority 2
switch(config)# map queue 3 local-priority 3
switch(config)# map queue 4 local-priority 4
switch(config)# map queue 5 local-priority 5
switch(config)# map queue 5 local-priority 6
switch(config)# map queue 5 local-priority 7
switch(config)# qos schedule-profile S1
switch(config)# wfq queue 0 weight 5
switch(config)# wfq queue 1 weight 10
switch(config)# wfq queue 3 weight 15
switch(config)# wfq queue 4 weight 25
switch(config)# wfq queue 5 weight 50
switch(config)# qos threshold-profile T1
switch(config-threshold)# queue 4 action ecn threshold 40 kbytes
switch(config-threshold)# queue 5 action ecn threshold 50 kbytes
switch(config-threshold)# exit
switch(config)# apply qos threshold-profile T!
```

```
switch(config)# wfq queue 5 weight 50
switch(config)# apply qos queue-profile Q1 schedule-profile S1
```

Configuring expedited forwarding for VoIP traffic

Voice over IP (VoIP) traffic is delay and jitter sensitive. Therefore, for optimum transmission of VoIP traffic, dwell time in network devices must be kept to a minimum, and all network devices in the data path must have identical per-hop behaviors. To configure a dedicated queue on the switch to handle VoIP traffic with priority service before all other queues, follow these steps.

Prerequisites

This scenario assumes that VoIP packets are uniquely identified using DiffServ code point 46, Expedited Forwarding (EF).

Procedure

1. Map DSCP EF packets exclusively to local priority 5. The default DSCP map has eight code points (40 through 47), that are mapped to local priority 5. To reserve local priority 5 for VoIP traffic, the other code points must be reassigned. In this scenario, local priority 6 is used for all reassignments, including for code point 40, Call Signaling protocol (CS5).

```
switch# config
switch(config)# qos dscp-map 40 local-priority 6 name CS5
switch(config)# qos dscp-map 41 local-priority 6
switch(config)# qos dscp-map 42 local-priority 6
switch(config)# qos dscp-map 43 local-priority 6
switch(config)# qos dscp-map 44 local-priority 6
switch(config)# qos dscp-map 45 local-priority 6
switch(config)# qos dscp-map 47 local-priority 6
```

2. Queue 7 is the highest priority queue, so for best throughput, create a queue profile that maps local priority 5 to queue 7.

```
switch(config)# qos queue-profile ef_priority
switch(config-queue)# name queue 7 Voice_Priority_Queue
switch(config-queue)# map queue 7 local-priority 5
switch(config-queue)# map queue 6 local-priority 7
switch(config-queue)# map queue 5 local-priority 6
switch(config-queue)# map queue 4 local-priority 4
switch(config-queue)# map queue 3 local-priority 3
switch(config-queue)# map queue 2 local-priority 2
switch(config-queue)# map queue 1 local-priority 1
switch(config-queue)# map queue 0 local-priority 0
switch(config-queue)# exit
switch(config)#
```

3. Create a schedule profile that services queue 7 using strict priority (SP), and the remaining queues with WFQ. This scenario gives all WFQ queues equal weight.

```
switch(config)# qos schedule-profile voip
switch(config-schedule)# strict queue 7
switch(config-schedule)# wfq queue 6 weight 1
switch(config-schedule)# wfq queue 5 weight 1
switch(config-schedule)# wfq queue 4 weight 1
switch(config-schedule)# wfq queue 3 weight 1
switch(config-schedule)# wfq queue 2 weight 1
switch(config-schedule)# wfq queue 1 weight 1
```

```
switch(config-schedule)# wfq queue 0 weight 1
switch(config-schedule)# exit
switch(config)#
```

4. Apply the profiles to all interfaces.

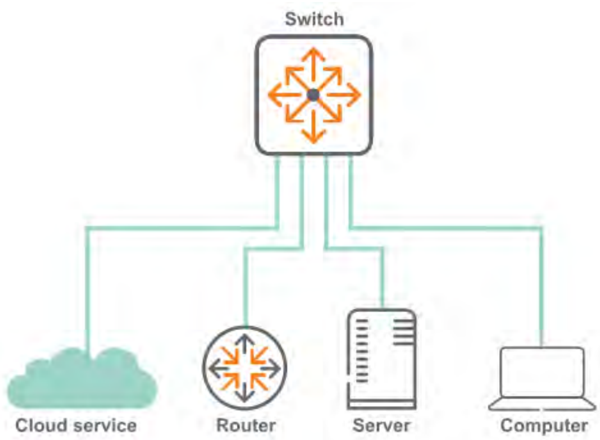
```
switch(config)# apply qos queue-profile ef_priority schedule-profile voip
```

5. Configure DSCP trust mode on all ports

```
switch(config)# qos trust dscp
```

Configuring rate limiting

This scenario illustrates how to use rate limiting to manage the traffic from various devices connected to a switch. The physical topology of the network looks like this:



A certain amount of broadcast traffic is necessary to maintain healthy network operation, particularly from routers and across service boundaries. In this scenario, both the service cloud and the router connections limit this traffic to 1 Gbps. The server has a smaller limit, as it does not require as much network protocol traffic as the service cloud and router.

A multicast server needs to be able to stream multicast traffic to clients, so a multicast rate limit may not be helpful. A computer, however, should not be generating large amounts of multicast traffic (it may be receiving streams, but typically not sending them). In this example, the computer is configured with a multicast rate limit to prevent malicious traffic from taking up network bandwidth.

Finally, while the service cloud and router may need to send traffic for unknown unicast addresses to resolve address forwarding, the server and computer should send very little of this type of traffic. Rate limiting unknown unicast traffic on those two devices enforces that.

Procedure

1. Configure broadcast and multicast rate limiting for the service cloud connection.

```
switch# config
switch(config)# interface 1/1/1
switch(config-if)# rate-limit broadcast 1000000 kbps
switch(config-if)# rate-limit multicast 2000000 kbps
switch(config-if)# exit
```

2. Configure broadcast rate limiting for the router connection.

```
switch(config-if)# interface 1/1/2
switch(config-if)# rate-limit broadcast 1000000 kbps
switch(config-if)# exit
```

3. Configure broadcast and unknown unicast rate limiting for the server connection.

```
switch(config-if)# interface 1/1/5
switch(config-if)# rate-limit broadcast 500000 kbps
switch(config-if)# rate-limit unknown-unicast 500 kbps
switch(config-if)# exit
```

4. Configure broadcast, unknown unicast, and multicast rate limiting for the computer connection.

```
switch(config-if)# interface 1/1/10
switch(config-if)# rate-limit broadcast 1000 kbps
switch(config-if)# rate-limit multicast 500 kbps
switch(config-if)# rate-limit unknown-unicast 200 kbps
```

Configuring egress queue shaping

This example shows how to apply egress queue shaping to an interface. First, a schedule profile is created that has per-queue bandwidth limits set on all queues with `strict` as the scheduling algorithm. Next, this profile is applied to an interface or LAG.

The following example creates a schedule profile named **`EQSExample`**, which services all eight queues using `strict` priority. This profile configures queues 1, 4, and 7 with a bandwidth limit of 10 Gbps, 20 Gbps, and 30 Gbps respectively. Also, queues 1 and 7 are configured with a burst of 120 KB (burst configuration is only supported on the 8320 and 8325). The profile is then applied to interface 1/1/1. (The actual burst and bandwidth configured on the interface can be found by using the `show interface <IFNAME> qos` command.)

```
switch(config)# qos schedule-profile EQSExample
switch(config-schedule)# strict queue 0
switch(config-schedule)# strict queue 1 max-bandwidth 10000000 burst 120
switch(config-schedule)# strict queue 2
switch(config-schedule)# strict queue 3
switch(config-schedule)# strict queue 4 max-bandwidth 20000000
switch(config-schedule)# strict queue 5
switch(config-schedule)# strict queue 6
switch(config-schedule)# strict queue 7 max-bandwidth 30000000 burst 120
switch(config-schedule)# exit
switch(config)# interface 1/1/1
switch(config-if)# apply qos schedule-profile EQSExample
```

Configuring egress port shaping

This example shows how to apply egress port shaping to an interface to limit the rate of egress traffic. Egress port shaping is configured by specifying the desired bandwidth rate in kilobits per second (kbps). An optional burst size in kilobytes (KB) may be configured for the Aruba 8400 switch. To be effective, the egress rate must be less than the line rate of the egress interface. If the configured egress rate exceeds the interface's line rate, then egress shaping has no effect.

The configured egress rate and burst on a specific interface can be found by using the `show interface` and `show interface qos` commands.

The following example configures an egress rate of 100 Mbps and a burst of 70 KB.

```
switch(config)# interface 1/1/1
switch(config-if)# qos shape 100000 burst 70
```

In the next example, both egress port shaping and egress queue shaping are configured on the same interface.

The example creates a schedule profile named `EQSExample` with strict priority for all seven queues. Queue 7 is configured with a bandwidth limit of 300 Mbps and a burst of 120 KB. The profile is then applied to interface `1/1/1` with egress port shaping of 400 Mbps and a burst size of 80 KB. As egress queue shaping and egress port shaping are both configured on port `1/1/1`, egress queue shaping is subject to the lower port or queue shape rate. The effective bandwidth for the traffic egressing on queue 7 will be 300 Mbps and the effective burst will be 80 KB.

```
switch(config)# qos schedule-profile EQSExample
switch(config-schedule)# strict queue 0
switch(config-schedule)# strict queue 1
switch(config-schedule)# strict queue 2
switch(config-schedule)# strict queue 3
switch(config-schedule)# strict queue 4
switch(config-schedule)# strict queue 5
switch(config-schedule)# strict queue 6
switch(config-schedule)# strict queue 7 max-bandwidth 300000 burst 120
switch(config-schedule)# exit
switch(config)# interface 1/1/1
switch(config-if)# apply qos schedule-profile EQSExample
switch(config-if)# qos shape 400000 burst 80
```

Supporting Ethernet 802.1D Class of Service

IEEE 802.1Q is the most current Ethernet standard for Class of Service (CoS). It superseded an earlier standard, 802.1D, in 2005. IEEE 802.1Q slightly changed the ordering of the classes of service from its predecessor IEEE 802.1D for CoS 2 and CoS 0:

CoS 802.1Q	CoS 802.1D
7 Network Control	7 Network Control
6 Internetwork Control	6 Voice (<10ms latency)
5 Voice (<10ms latency)	5 Video (<100ms latency)
4 Video (<100ms latency)	4 Controlled Load
3 Critical Applications	3 Excellent Effort
2 Excellent Effort	0 Best Effort
0 Best Effort	2 Spare
1 Background	1 Background

In 802.1D, both CoS 2 and CoS 1 are below CoS 0 (Best Effort).

When a switch is installed in a network of devices following 802.1D Class of Service, the QoS Cos map must be re-configured to follow the 802.1D standard by swapping the assignments of CoS 0 and 2:

```
switch# config
switch(config)# qos cos-map 0 local-priority 2 color green name Best_Effort
switch(config)# qos cos-map 2 local-priority 1 color green name Spare
switch(config)# exit
switch# show qos cos-map
code_point local_priority color   name
---------- -------------- ------- ----
0          2              green   Best_Effort
1          0              green   Background
2          1              green   Spare
```

```
3           3              green   Critical_Applications
4           4              green   Video
5           5              green   Voice
6           6              green   Internetwork_Control
7           7              green   Network_Control
```

Monitoring queue operation

Use the show interface queues command to display the traffic transmitted per queue, and the number of packets dropped due to the queue being full.. For example:

```
switch# show interface 1/1/5 queues
Interface 1/1/5 is up
 Admin state is up
              Tx Bytes      Tx Packets     Tx Drops     Tx Byte Depth
 Q0        157113373520      1890863919            0              1362
 Q1        233312143017      2808451320           18             65550
 Q2        156814056423      1887257650            0              1392
 Q3        157441358980      1894815504            0              1374
 Q4        157700809294      1897941370            0              1362
 Q5        157872849381      1900014146            0              1392
 Q6        183486049854      2208268429            0              4398
 Q7        231607534141      2787913734            0             65544
```

- **Tx Bytes**: Total bytes transmitted. The byte count may include packet headers and internal metadata that are removed before the packet is transmitted. Packet headers added when the packet is transmitted may not be included.
- **Tx Packets**: Total packets transmitted.
- **Tx Drops**: The number of packets dropped by a queue before it was sent. When traffic cannot be forwarded out an egress interface, it backs up at ingress. The more servicing assigned to a queue by a schedule profile, the less likely traffic destined for that queue will back up and be dropped. Tx Drops shows the sum of packets that were dropped across all line modules (due to insufficient capacity) by the ingress Virtual Output Queues (VOQs) destined for the egress port. As the counts are read separately from each line module, the sum is not an instantaneous snapshot.
- Tx Byte Depth: Largest byte depth (or high watermark) found on any ingress line module Virtual Output Queue (VOQ) destined for the egress port.

apply qos

Syntax

```
apply qos [queue-profile <QUEUE-NAME>] schedule-profile <SCHEDULE-NAME>

no apply qos schedule-profile <SCHEDULE-NAME>
```

Description

Applies a queue profile and schedule profile globally to all Ethernet and LAG interfaces on the switch, or applies a schedule profile to a specific interface. When applied globally, the specified schedule profile is configured only on Ethernet interfaces and LAGs that do not already have their own schedule profile.

The same profile can be applied both globally and locally to an interface. This guarantees that an interface always uses the specified profile, even if the global profile is changed.

The `no` form of this command removes the specified schedule profile from an interface and the interface uses the global schedule profile. This is the only way to remove a schedule profile override from the interface.

NOTE: When applying QoS settings to a port configured to support priority-based flow control, specific configuration settings must be respected when defining a CoS map and queue profile. See the command `flow-control` in the *ArubaOS-CX Command-Line Interface Guide* for details.

NOTE: Interfaces may shut down briefly during reconfiguration.

Command context

`config`

`config-if`

`config-lag-if`

Parameters

`queue-profile <QUEUE-NAME>`

Specifies the name of the queue profile to apply. Range: 1 to 64 alphanumeric characters, including period (.), underscore (_), and hyphen (-). This parameter is not supported in the `config-if` context.

`schedule-profile <SCHEDULE-NAME>`

Specifies the name of the schedule profile to apply. Range: 1 to 64 alphanumeric characters, including period (.), underscore (_), and hyphen (-).

Authority

Administrators or local user group members with execution rights for this command.

Usage

- For a queue profile to be complete and ready to be applied, all eight local priorities must be mapped to a queue.
- For a schedule profile to be complete and ready to be applied, it must define all queues specified in the queue profile. All queues must use the same algorithm, except for the highest numbered queue, which can be **`strict`**.
- Both the queue profile and the schedule profile must specify the same number of queues.
- Schedule profiles can be modified while applied, but only in ways where a single command will not result in the profile becoming invalid. For example, queue 7 can have the algorithm changed, and weighted queues can have their weights changed.
- The following commands illustrate a valid configuration, where every local priority value is assigned to a queue, and all assigned queues are defined.

```
map queue 0 local-priority 0
map queue 1 local-priority 1
map queue 1 local-priority 2
map queue 3 local-priority 3
map queue 4 local-priority 4
map queue 5 local-priority 5
map queue 5 local-priority 6
map queue 5 local-priority 7

wfq queue 0 weight 5
wfq queue 1 weight 10
wfq queue 3 weight 15
wfq queue 4 weight 25
wfq queue 5 weight 50
```

The following commands illustrate an invalid configuration, because local priority 2 is not assigned to a queue.

```
qos cos-map 1 local-priority 1
qos queue-profile Q1
map queue 0 local-priority 0
map queue 1 local-priority 1
map queue 3 local-priority 3
map queue 4 local-priority 4
map queue 5 local-priority 5
map queue 5 local-priority 6
map queue 5 local-priority 7

wfq queue 0 weight 5
wfq queue 1 weight 10
wfq queue 3 weight 15
wfq queue 4 weight 25
wfq queue 5 weight 50
```

Example

Applying the QoS profile **`myprofile`** and the schedule profile **`myschedule`** to all interfaces that do not have an applied interface-specific schedule profile:

```
switch(config)# apply qos queue-profile myprofile schedule-profile myschedule
```

apply qos threshold-profile

Syntax

```
apply qos <THRESHOLD-NAME>
```

```
no apply qos schedule-profile
```

Description

Applies a threshold profile globally to all Ethernet and LAG interfaces on the switch, or to a specific interface. When applied globally, the specified threshold profile is configured only on Ethernet interfaces and LAGs that do not already have their own schedule profile.

The same profile can be applied both globally and locally to an interface. This guarantees that an interface always uses the specified threshold profile, even if the global profile is changed.

The `no` form of this command removes the specified threshold profile from an interface, and causes it to use the global threshold profile. This is the only way to remove a threshold profile override from an interface. A profile can only be deleted once it is no longer applied to any interface.

Command context

`config`

`config-if`

`config-lag-if`

Parameters

<THRESHOLD-NAME>

Specifies the name of the threshold profile to apply. Range: 1 to 64 alphanumeric characters, including period (.), underscore (_), and hyphen (-).

Authority

Administrators or local user group members with execution rights for this command.

Example

Applying the threshold profile **mythreshold** to all interfaces that do not have an applied profile:

```
switch(config)# apply qos threshold-profile mythreshold
```

map queue

Syntax

```
map queue <QUEUE-NUMBER> local-priority <PRIORITY-NUMBER>
```

```
no map queue <QUEUE-NUMBER> [local-priority <PRIORITY-NUMBER>]
```

Description

Assigns a local priority to a queue in a queue profile. By default, the larger the queue number the higher its priority. A queue without a local priority value assigned to it is not used to store packets. The same queue can be assigned multiple local priorities.

The `no` form of this command removes the specified local priority from a specific queue. If no local priority number is specified, then all local priorities are removed from the queue.

Command context

`config-queue`

Parameters

<QUEUE-NUMBER>

Specifies the queue number. Range: 0 to 7.

<PRIORITY-NUMBER>

Specifies the queue priority. Range: 0 to 7, where 0 is the lowest priority and 7 is the highest.

Authority

Administrators or local user group members with execution rights for this command.

Usage

For a queue profile to be complete and ready to be applied, all eight local priorities must be mapped to a queue.

The following commands illustrate an invalid configuration, where every local priority value is assigned to a queue.

```
map queue 0 local-priority 0
map queue 1 local-priority 1
map queue 1 local-priority 2
map queue 3 local-priority 3
map queue 4 local-priority 4
map queue 5 local-priority 5
map queue 5 local-priority 6
map queue 5 local-priority 7
```

The following commands illustrate an invalid configuration, because local priority 2 is not assigned to a queue.

```
map queue 0 local-priority 0
map queue 1 local-priority 1
map queue 2 local-priority 3
map queue 3 local-priority 4
map queue 4 local-priority 5
map queue 5 local-priority 6
map queue 5 local-priority 7
```

Examples

Assigning priority **7** to queue **7** in profile **myprofile**:

```
switch(config)# qos queue-profile myprofile
switch(config-queue)# map queue 7 local-priority 7
```

Removing priority **7** from queue **7** in profile **myprofile**:

```
switch(config)# qos queue-profile myprofile
switch(config-queue)# no map queue 7 local-priority 7
```

name queue

Syntax

```
name queue  <QUEUE-NUMBER> <DESCRIPTION>

no name queue  <QUEUE-NUMBER>
```

Description

Assigns a description to a queue in a queue profile. This is for identification purposes and has no effect on configuration.

The `no` form of this command removes the description associated with a queue.

Command context

`config-queue`

Parameters

<QUEUE-NUMBER>

Specifies the queue number. Range: 0 to 7.

<DESCRIPTION>

Specifies a queue description for identification purposes. Range: 1 to 64 alphanumeric characters, including period (.), underscore (_), and hyphen (-).

Authority

Administrators or local user group members with execution rights for this command.

Examples

Assigning the description **`priority-traffic`** to queue **7**:

```
switch(config)# qos queue-profile myprofile
switch(config-queue)# name queue 7 priority-traffic
```

Removing the description from queue **7**:

```
switch(config)# qos queue-profile myprofile
switch(config-queue)# no name queue 7
```

qos cos-map

Syntax

```
qos cos-map <CODE-POINT> local-priority <PRIORITY-NUMBER> [color <COLOR>] [name <DESCRIPTION>]

no qos cos-map <CODE-POINT>
```

Description

Defines the local priority assigned to incoming packets for a specific 802.1 VLAN priority code point (CoS) value. The CoS map values are used to mark incoming packets when QoS trust mode is set to **`cos`**. In **trust none** mode, CoS map entry 0 is used to set the port default local priority and color.

To see the default CoS map settings, use the following command:

```
switch# show qos cos-map default
code_point local_priority color   name
---------- -------------- ------- ----
0          1              green   Best_Effort
1          0              green   Background
2          2              green   Excellent_Effort
3          3              green   Critical_Applications
4          4              green   Video
5          5              green   Voice
6          6              green   Internetwork_Control
7          7              green   Network_Control
```

The `no` form of this command restores the assignments for a CoS map value to the default setting.

Command context

`config`

Parameters

<CODE-POINT>

Specifies an 802.1 VLAN priority CoS value. Range: 0 to 7. Default 0.

local-priority *<PRIORITY-NUMBER>*

Specifies a local priority value to associate with the `CODE-POINT` value. Range: 0 to 7. Default: 0.

color <COLOR>

Reserved for future use.

name *<DESCRIPTION>*

Specifies a description for the CoS setting. The name is for identification only, and has no effect on queue configuration. Range: 1 to 64 alphanumeric characters, including period (.), underscore (_), and hyphen (-).

Authority

Administrators or local user group members with execution rights for this command.

Examples

Mapping CoS value **1** to a local priority of **2**:

```
switch(config)# qos cos-map 1 local-priority 2
```

Mapping CoS value **1** to the default local priority value:

```
switch(config)# no qos cos-map 1
```

qos dscp-map

Syntax

`qos dscp-map <CODE-POINT> local-priority <PRIORITY-NUMBER> [color <COLOR>] [cos COS-CODE-POINT] [name <DESCRIPTION>]`

`no qos dscp-map <CODE-POINT>`

Description

Defines the local priority assigned to incoming packets for a specific IP differentiated services code point (DSCP) value. The DSCP map values are used to prioritize incoming packets when QoS trust mode is set to **dscp**.

To see the default DSCP map settings, use the following command:

```
switch# show qos dscp-map default
code_point local_priority cos color   name
---------- -------------- --- ------- ----
0          1                  green   CS0
1          1                  green
2          1                  green
3          1                  green
4          1                  green
5          1                  green
...
45         5                  green
46         5                  green   EF
47         5                  green
48         6                  green   CS6
...
61         7                  green
62         7                  green
63         7                  green
```

The `no` form of this command restores the assignments for a code point to the default setting.

Command context

`config`

Parameters

<CODE-POINT>

Specifies an IP differentiated services code point. Range: 0 to 63. Default: 0.

local-priority *<PRIORITY-NUMBER>*

Specifies a local priority value to associate with the `CODE-POINT` value. Range: 0 to 7. Default: 0.

color *<COLOR>*

Reserved for future use.

cos *<COS-CODE-POINT>*

Specifies an 802.1 VLAN priority CoS remark value. Range: 0 to 7. Default 0

name *<DESCRIPTION>*

Specifies a description for the DSCP setting. The name is used for identification only, and has no effect on queue configuration. Range: 1 to 64 alphanumeric characters, including period (.), underscore (_), and hyphen (-).

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting code point **1** to a local priority of **2** and a CoS of **0**:

```
switch(config)# qos dscp-map 1 local-priority 2 cos 0
```

Setting code point **1** to the default value:

```
switch(config)# no qos dscp-map 1
```

qos queue-profile

Syntax

```
qos queue-profile <NAME>

no qos queue-profile <NAME>
```

Description

Creates a new QoS queue profile and switches to the `config-queue` context for the profile. Or, if the specified QoS queue profile exists, this command switches to the `config-queue` context for the profile.

A queue profile maps queues to local-priority values.

Each profile has eight queues, numbered 0 to 7. The larger the queue number, the higher its priority during transmission scheduling.

A queue profile named **`factory-default`** is defined by default and is automatically applied to all interfaces. It cannot be edited or deleted. Use the command `show qos queue-profile factory-default` to view this profile. Do not use `show running-configuration`, as it will only display changes from the initial values.

The `no` form of this command removes the specified QoS queue profile. Only profiles that are not currently applied can be removed.

Command context

`config`

Parameters

<NAME>

Specifies the name of the QoS queue profile to create or configure. Range: 1 to 64 alphanumeric characters, including period (.), underscore (_), and hyphen (-).

Authority

Administrators or local user group members with execution rights for this command.

Examples

Creating the profile **`myprofile`**:

```
switch(config)# qos queue-profile myprofile
switch(config-queue)#
```

Deleting the profile **`myprofile`**:

```
switch(config)# no qos queue-profile myprofile
```

qos schedule-profile

Syntax

```
qos schedule-profile <NAME>

no qos schedule-profile <NAME>
```

Description

Creates a QoS schedule profile and switches to the `config-schedule` context for the profile. If the specified schedule profile exists, this command switches to the `config-schedule` context for the profile. The schedule profile determines the order in which queues are selected to transmit a packet, and the amount of service defined for each queue.

Queues in a schedule profile are numbered consecutively starting from zero. Queue zero is the lowest priority queue. The larger the queue number, the higher priority the queue has in scheduling algorithms.

A profile named **`factory-default`** is defined by default and applied to all interfaces. It cannot be edited or deleted. To see its settings, use the command:

```
switch# show qos schedule-profile factory-default
queue_num algorithm weight
--------- --------- ------
0         wfq       1
1         wfq       1
2         wfq       1
3         wfq       1
4         wfq       1
5         wfq       1
6         wfq       1
7         wfq       1
```

A profile named **`strict`** is predefined and cannot be edited or deleted. The strict profile services all queues of the queue profile to which it is applied, using the strict priority algorithm.

A schedule profile must be defined on all interfaces at all times.

There are two permitted configurations for a schedule profile:

1. All queues use the same scheduling algorithm (for example, WFQ).
2. The highest queue number uses strict priority, and all remaining (lower) queues use the same algorithm (for example, WFQ). This supports priority scheduling behavior necessary for the IEFT RFC 3246 Expedited Forwarding specification (https://tools.ietf.org/html/rfc3246).

Only limited changes can be made to an applied schedule profile:

- The weight of a wfq queue.
- The bandwidth of a strict queue.
- The algorithm of the highest numbered queue can be swapped between wfq and strict, and vice versa.

Applicable to REST: Any other changes will result in an unusable schedule profile, and the switch will revert to the `factory-default` profile until the profile is corrected.

The `no` form of this command removes the specified QoS schedule profile when it is not applied. Only profiles that are not currently applied to an interface can be removed.

Command context

`config`

Parameters

<NAME>

Specifies the name of the QoS schedule profile to create or configure. Range: 1 to 64 alphanumeric characters, including period (.), underscore (_), and hyphen (-).

Authority

Administrators or local user group members with execution rights for this command.

Examples

Creating the schedule profile **myschedule**:

```
switch(config)# qos schedule-profile myschedule
switch(config-schedule)#
```

Deleting the schedule profile **myschedule**:

```
switch(config)# no qos schedule-profile myschedule
```

qos shape

Syntax

qos shape *<RATE>* [burst *<SIZE>*]

no qos shape

Description

Limits the egress bandwidth on an interface to a value that is lower than its line rate. An optional 'burst' value may also be specified.

The no form of this command removes shaping from an interface.

Command context

config-if

Parameters

<RATE>

Specifies the maximum traffic rate in kbps. Range: 468 to100000000.

<SIZE>

Specifies the maximum burst size in kilobytes. Range: 0 to 127.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring an egress port shaping rate of 400 Mbps on interface **1/1/1** with a burst size of 50 KB:

```
switch(config)# interface 1/1/1
switch(config-if)# qos shape 400000 burst 50
```

Deleting egress port shaping on interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# no qos shape
```

qos threshold-profile

Syntax

```
qos threshold-profile <NAME>

no qos threshold-profile <NAME>
```

Description

Creates a QoS threshold profile and switches to the `config-threshold` context for the profile. If the specified threshold profile exists, this command switches to the `config-threshold` context for the existing profile. The threshold profile determines the action to take when a threshold is exceeded for each queue.

A threshold profile is composed of up to 8 queues, numbered from 0 to 7. Each queue defines the action to take when buffer utilization exceeds a specific threshold. Configure queues with the command `queue`,

The `no` form of this command removes the specified QoS threshold profile. Only profiles that are not currently applied to an interface can be removed.

Command context

`config`

Parameters

<NAME>

Specifies the name of the QoS threshold profile to create or configure. Range: 1 to 64 alphanumeric characters, including period (.), underscore (_), and hyphen (-).

Authority

Administrators or local user group members with execution rights for this command.

Examples

Creating the threshold profile **`mythreshold`**:

```
switch(config)# qos threshold-profile mythreshold
switch(config-threshold)#
```

Deleting the threshold profile **`myschedule`**:

```
switch(config)# no qos threshold-profile mythreshold
```

qos trust

Syntax

```
qos trust {none | cos | dscp}

no qos trust
```

Description

Sets the trust mode. Trust mode determines whether CoS or DSCP values are used to assign local priority values to ingress packets. CoS values are taken from the CoS map, and DSCP values are taken from the DSCP map.

In the `config` context:

- This command sets the trust mode that is globally applied to all interfaces that do not have a trust mode configured.
- The `no` form of this command restores all interfaces that do not currently have a trust mode configured to the default setting (`none`).

In the `config-if` context:

- This command sets the trust mode override for a specific interface.
- The `no` form of this command clears a trust mode override. The interface then uses the global setting. This is the only way to remove a trust mode override.

Command context

`config`

`config-if`

Parameters

none

Ignore all packet headers. Ingress packets are assigned the priority value configured for CoS map value `0`. Default.

cos

For 802.1 VLAN-tagged packets, use the priority code point field value from the outermost VLAN header as the index into the CoS map. If the packet is untagged, use the priority configured for CoS map value `0`.

dscp

For IP packets, use the DSCP value as the index into the DSCP map. For non-IP packets, use the priority configured for CoS map value `0`.

Authority

Administrators or local user group members with execution rights for this command.

Example

Setting the global trust mode to **dscp**, which is applied to all interfaces that do not already have an individual trust mode configured. An override is then applied to interface **2/2/2**, and LAG 100, setting trust mode to **cos**:

```
switch(config)# qos trust dscp
switch(config)# interface 2/2/2
switch(config-if)# qos trust cos
switch(config-if)# interface lag 100
switch(config-if)# qos trust cos
```

queue

Syntax

`queue <QUEUE-NUMBER> action ecn all threshold  <AMOUNT> kbytes`

`no queue <QUEUE-NUMBER> action ecn all threshold  <AMOUNT> kbytes`

Description

Defines the threshold value and action for a queue in a threshold-profile. When queue utilization exceeds the threshold value, ECT (ECN-Capable Transport) packets will be CE (Congestion Encountered) marked when transmitted.

The `no` form of this command removes the settings for a queue.

Command context

`config-threshold`

Parameters

<QUEUE-NUMBER>

Specifies the queue number. Range: 0 to 7.

`action ecn`

Apply ECN when the threshold is exceeded.

`all`

Applies the action to all colors. Colors are reserved for future use.

`threshold` ***`<AMOUNT>`*** **`kbytes`**

Specifies the threshold value in kbytes from 0 to 1700.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Assigning a threshold of **`40`** kilobytes to queue **`7`** in profile **`mythreshold`**:

```
switch(config)# qos threshold-profile mythreshold
switch(config-threshold)# queue 7 action ecn all threshold 40 kbytes
```

Removing a threshold of **`40`** kilobytes from queue **`7`** in profile **`mythreshold`**:

```
switch(config)# qos threshold-profile mythreshold
switch(config-threshold)# no queue 7
```

rate-limit

Syntax

```
rate-limit {broadcast | multicast | unknown-unicast} <RATE> {kbps | pps}

no rate-limit {broadcast | multicast | unknown-unicast}
```

Description

Sets the amount of traffic of a specific type that can ingress on an Ethernet port, or on each port of a LAG interface. Rate limits are enforced separately on each individual member of a LAG, not on the LAG as a whole.

The `no` form of this command removes the traffic limit for the specified traffic type.

Command context

`config-if`

Parameters

{broadcast | multicast | unknown-unicast}

Specifies the type of ingress traffic to which the rate limit applies: `broadcast`, `multicast`, or `unknown-unicast`. The multicast rate limit affects multicast and broadcast traffic. The broadcast rate limit only affects broadcast traffic. When both types are applied to the same interface, broadcast packets are limited to the lower of the two rate values. Layer 2 BPDU packets, like spanning tree, are also included in the multicast rate limit.

***<RATE>* {kbps | pps}**

Specifies the rate limit. Range: 22 to 100000000 kbps (in steps of 22 kbps), or 43 to 209090910 kbps (in steps of 43 pps). The actual rate limit varies with steps approximately equal to the minimum value. Verify the actual rate limit using the command `show interface <INTERFACE-NAME>`.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Limiting **broadcast** traffic to **500 kbps** on interface **1/1/3**:

```
switch(config)# interface 1/1/3
switch(config-if)# rate-limit broadcast 500 kbps
```

Limiting **multicast** traffic to **4000 pps** on interface **1/1/3**:

```
switch(config)# interface 1/1/3
switch(config-if)# rate-limit multicast 4000 pps
```

Limiting **unknown unicast** traffic to **100 kbps** on interface **1/1/3**.

```
switch(config)# interface 1/1/3
switch(config-if)# rate-limit unknown-unicast 100 kbps
```

Viewing the results of the previous configuration settings.

```
switch# show interface 1/1/3 qos
Interface 1/1/3 is down (Administratively down)
 Admin state is down
 Hardware: Ethernet, MAC Address: 1c:98:ec:e3:6a:00
 MTU 1500
 Full-duplex
 rate-limit broadcast unknown-unicast 100 kbps (109 actual)
 rate-limit broadcast 500 kbps (505 actual)
 rate-limit multicast 4000 pps (3990 actual)

 Speed 0 Mb/s
 Auto-Negotiation is turned on
 Input flow-control is off, output flow-control is off
 RX
            0 input packets              0 bytes
            0 input error                0 dropped
            0 CRC/FCS
       L3:
            ucast: 0 packets, 0 bytes
            mcast: 0 packets, 0 bytes
 TX
            0 output packets             0 bytes
            0 input error                0 dropped
            0 collision
       L3:
```

```
            ucast: 0 packets, 0 bytes
            mcast: 0 packets, 0 bytes
```

Limiting **broadcast** traffic to **50 kbps** on LAG **100**.

```
switch# config
switch(config)# interface 1/1/3
switch(config)# interface lag 100
switch(config-if)# rate-limit broadcast 50 kbps
```

show interface queues

Syntax

`show interface <INTERFACE-NAME> queues [vsx-peer]`

Description

Shows the traffic transmitted per queue and the number of packets dropped due to the queue being full.

NOTE: For every layer 3 packet transmitted on a switch queue, an extra three bytes are added to the value of Tx Bytes (due to the inclusion of an internal routing packet header). To show layer 3 statistics without this addition, enable layer 3 counters on an interface, and execute the command `show interface`. For example:

```
switch(config)# interface 1/1/1
switch(config-if)# l3-counters
switch(config-if)# exit
switch# show interface 1/1/1
```

Command context

Operator (>) or Manager (#)

Parameters

<INTERFACE-NAME>

Specifies the name of an interface on the switch. Format: `member/slot/port`.

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Usage

Statistics include:

- **Tx Bytes:** Total bytes transmitted. The byte count may include packet headers and internal metadata that are removed before the packet is transmitted. Packet headers added when the packet is transmitted may not be included.
- **Tx Packets:** Total packets transmitted.

- **Tx Drops:** The number of packets dropped by a queue before it was sent. When traffic cannot be forwarded out an egress interface, it backs up at ingress. The more servicing assigned to a queue by a schedule profile, the less likely traffic destined for that queue will back up and be dropped. Tx Drops shows the sum of packets that were dropped across all line modules (due to insufficient capacity) by the ingress Virtual Output Queues (VOQs) destined for the egress port.
- **Tx Byte Depth:** Largest byte depth (or high watermark) found on any ingress line module Virtual Output Queue (VOQ) destined for the egress port.

Examples

Showing settings for interface `1/1/5`:

```
switch# show interface 5
Interface 1/1/5 is down (Administratively down)
 Admin state is down
 State information: admin_down
 Hardware: Ethernet, MAC Address: aa:55:aa:55:00:29
 MTU 1500
 Full-duplex
 qos trust cos
 qos schedule-profile default
 qos shape 400000 burst 70
 rate-limit broadcast 500 kbps
 rate-limit multicast 4000 pps
 rate-limit broadcast unknown-unicast 100 kbps
 Speed 0 Mb/s
 Auto-Negotiation is turned on
 Flow-control: off
```

Showing queue statistics for interface `1/1/5`:

```
switch# show interface 1/1/5 queues
Interface 1/1/5 is up
 Admin state is up
               Tx Bytes       Tx Packets          Tx Drops           Tx Byte Depth
 Q0         157113373520      1890863919                 0                    1362
 Q1         233312143017      2808451320                18                   65550
 Q2         156814056423      1887257650                 0                    1392
 Q3         157441358980      1894815504                 0                    1374
 Q4         157700809294      1897941370                 0                    1362
 Q5         157872849381      1900014146                 0                    1392
 Q6         183486049854      2208268429                 0                    4398
 Q7         231607534141      2787913734                 0                   65544
```

show interface qos

Syntax

`show interface <INTERFACE-NAME> qos`

Description

Shows various QoS settings for a specific interface.

Command context

Operator (>) or Manager (#)

Parameters

<INTERFACE-NAME>

Specifies the name of an interface on the switch. Format: `member/slot/port or lag number`.

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing QoS settings for interface **1/1/5**:

```
switch# show interface 1/1/5 qos
Interface 1/1/5 is down
 Admin state is up
 qos trust none (global)
 qos queue-profile factory-default (global)
 qos schedule-profile EQSExample (override)
 qos threshold-profile DefaultThresh (global)
 rate-limit unknown-unicast 50 pps (41 actual)
 rate-limit broadcast 500 kbps (505 actual)
 rate-limit multicast 500 kbps (505 actual)

         Max-Bandwidth Kbps      Burst KB
 Q1               10082461          120
 Q4               20164923           32
 Q7               30247384          120
```

show qos cos-map

Syntax

`show qos cos-map [default] [vsx-peer]`

Description

Shows the global QoS CoS code point settings, or the factory default settings.

Command context

Operator (>) or Manager (#)

Parameters

default

Shows the factory default CoS code point settings.

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing the current CoS map:

```
switch# show qos cos-map
code_point local_priority color   name
---------- -------------- ------- ----
0          2              green   Bulk
1          0              green   Scavenger
2          1              green   Best_Effort
3          3              green   Critical_Applications
4          4              green   Video
5          5              green   Voice
6          6              green   Internetwork_Control
7          7              green   Network_Control
```

Showing the default CoS map:

```
switch# show qos cos-map default
code_point local_priority color   name
---------- -------------- ------- ----
0          1              green   Best_Effort
1          0              green   Background
2          2              green   Excellent_Effort
3          3              green   Critical_Applications
4          4              green   Video
5          5              green   Voice
6          6              green   Internetwork_Control
7          7              green   Network_Control
```

(Color is reserved for future use.)

show qos dscp-map

Syntax

`show qos dscp-map [default] [vsx-peer]`

Description

Shows the global QoS DSCP code point settings, or the factory default settings.

Command context

Operator (>) or Manager (#)

Parameters

default

Shows the factory default DSCP code point settings.

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing the current DSCP map:

```
switch# show qos dscp-map
code_point local_priority color   name
---------- -------------- ------- ----
0          1              green   CS0
1          1              green
2          1              green
3          1              green
4          1              green
5          1              green
6          1              green
7          1              green
8          0              green   CS1
...
45         5              green
46         7              green   EF
47         5              green
48         6              green   CS6
...
61         7              green
62         7              green
63         7              green
```

Showing the default DSCP map:

```
switch# show qos dscp-map default
code_point local_priority color   name
---------- -------------- ------- ----
0          1              green   CS0
1          1              green
2          1              green
3          1              green
4          1              green
5          1              green
...
45         5              green
46         5              green   EF
47         5              green
48         6              green   CS6
...
61         7              green
62         7              green
63         7              green
```

(Color is reserved for future use.)

show qos queue-profile

Syntax

```
show qos queue-profile [<NAME> | factory-default ] [vsx-peer]
```

Description

Shows the status of all queue profiles, or a specific queue profile.

Command context

Operator (>) or Manager (#)

Parameters

<NAME>

Specifies the name of a queue profile. Range 1 to 64 alphanumeric characters, including period (.), underscore (_), and hyphen (-).

`factory-default`

Specifies the default profile.

`[vsx-peer]`

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Showing the status of queue profile **`myprofile`**:

```
switch# show qos queue-profile myprofile
queue_num local_priorities name
--------- ---------------- ----
0         2                Normal_and_Bulk
1         0                Other
2         1                Best_Effort
3         3
4         4
5         5
6         6
7         7                High_Priority
```

Showing the status of the default queue profile **`factory-default`**:

```
switch# show qos queue-profile factory-default
queue_num local_priorities name
--------- ---------------- ----
0         0                Scavenger_and_backup_data
1         1
2         2
3         3
4         4
5         5
6         6
7         7
```

show qos schedule-profile

Syntax

```
show qos schedule-profile [<NAME> | factory-default] [vsx-peer]
```

Description

Shows the status of all schedule profiles, or a specific schedule profile.

Command context

Operator (>) or Manager (#)

Parameters

<NAME>

Specifies the name of a queue or schedule profile. Range: 1 to 64 alphanumeric characters, including period (.), underscore (_), and hyphen (-).

factory-default

Specifies the default schedule profile.

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Showing the status of schedule profile **myschedule**:

```
switch# show qos schedule-profile myschedule
queue_num algorithm weight
--------- --------- ------
0         wfq       1
1         wfq       2
2         wfq       3
3         wfq       4
4         wfq       5
5         wfq       6
6         wfq       7
7         wfq       8
```

Showing the status of the default schedule profile:

```
switch# show qos schedule-profile factory-default
queue_num algorithm weight
--------- --------- ------
0         wfq       1
1         wfq       1
2         wfq       1
3         wfq       1
4         wfq       1
5         wfq       1
```

```
6          wfq        1
7          wfq        1
```

show qos threshold-profile

Syntax

`show qos threshold-profile [<NAME>] [vsx-peer]`

Description

Shows the status of all threshold profiles, or a specific threshold profile.

Command context

Operator (>) or Manager (#)

Parameters

<NAME>

Specifies the name of a threshold profile. Range: 1 to 64 alphanumeric characters, including period (.), underscore (_), and hyphen (-).

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Usage

Status field values are:

- applied : The threshold profile is applied to all configured ports.
- partially applied: The threshold profile is applied to some configured ports but failed on other configured ports.
- not applied : The threshold profile could not be applied to any configured ports.
- blank field: The threshold profile is not applied in the configuration (globally or as a port override).
- Error: Insufficient TCAM Resources: The switch hardware failed to activate a threshold profile.

Examples

Showing the status of threshold profile **mythreshold**:

```
switch# show qos threshold-profile mythreshold
queue_num action  Threshold
--------- ------- ---------
5         ecn       40
7         ecn       50

Ports       Status
---------   --------------------
```

```
1/1/1        applied
1/1/8        applied
```

Showing the status of threshold profile **CustomThresh** that failed to be activated by the switch hardware:

```
switch(config-if)# apply qos threhold-profile CustomThresh
switch(config-if)# show qos threshold-profile CustomThresh
queue_num action  Threshold
--------- ------- ---------
5         ecn       40
7         ecn       50

Ports        Status
---------    --------------------
1/2/2        Error: Insufficient TCAM Resources
```

show qos trust

Syntax

```
show qos trust [default] [vsx-peer]
```

Description

Shows the global QoS trust settings, or the factory default settings.

Command context

Operator (>) or Manager (#)

Parameters

default

Shows the factory default QoS trust settings.

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing the current QoS trust settings:

```
switch# show qos trust
qos trust cos
```

Showing the default QoS trust settings:

```
switch# show qos trust default
qos trust none
```

strict queue

Syntax

```
strict queue <QUEUE-NUMBER> [[max-bandwidth <BANDWIDTH>] [burst <BURST>]]

no strict queue <QUEUE-NUMBER>
```

Description

Assigns the strict priority algorithm to a queue. Strict priority services all packets waiting in a queue, before servicing the packets in lower priority queues.

Egress queue shaping can be configured using the `max-bandwidth` and `burst` options to limit the amount of traffic transmitted per output queue. The buffer associated with each egress queue stores the excess traffic to absorb bursts and smooth the output rate. Sustained rates of traffic above the maximum bandwidth will eventually fill the output queue causing tail drops. Use the command `show interface` to determine if any tail drop errors have occurred.

The `no` form of this command removes the the queue configuration from the schedule profile. To remove only egress queue shaping, re-enter the strict queue command without the `max-bandwidth` and `burst` parameters.

Command context

`config-schedule`

Parameters

<QUEUE-NUMBER>

Specifies the number of the queue. Range: 0 to 7.

max-bandwidth *<BANDWIDTH>*

Specifies the maximum bandwidth allowed on the queue in Kbps. Range: 1 to 4294967295.

burst *<BURST>*

Specifies the maximum burst allowed on the queue. Range: 1 to 127 Kbps. Default: 32 Kbps.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Assigning strict priority to queue **7** in the schedule profile **myschedule**:

```
switch(config)# qos schedule-profile myschedule
switch(config-schedule)# strict queue 7
```

Deleting strict priority from queue **7** in the schedule profile **myschedule**:

```
switch(config)# qos schedule-profile myschedule
switch(config-schedule)# no strict queue 7
```

Assigning strict priority to queue **7** in the schedule profile **myschedule** with a maximum bandwidth of **10000** Kbps and a burst of **62** Kbps :

```
switch(config)# qos schedule-profile myschedule
switch(config-schedule)# strict queue 7 max-bandwith 10000 burst 62
```

Assigning strict priority to queue **7** in the schedule profile **myschedule** with a maximum bandwidth of **10000** Kbps:

wfq queue

Syntax

```
wfq queue <QUEUE-NUMBER> weight <WEIGHT>

no wfq queue <QUEUE-NUMBER>
```

Description

Assigns the weighted fair queuing (WFQ) algorithm and its weight to a queue. Weighted fair queuing allocates available bandwidth among all queues that are not empty, in relation to their queue weights. WFQ is applied in bytes, not packets. It is work conserving, which means that only non-empty queues are counted on each scheduling cycle. The percentage of bandwidth allotted to a non-empty queue is calculated by dividing its weight by the sum of the weights for all non-empty queues. This means that the percentage of bandwidth allotted to a queue can fluctuate, depending on the number of non-empty queues present in each cycle.

The `no` form of this command removes the weighted fair queuing algorithm from a queue.

Command context

`config-schedule`

Parameters

<QUEUE-NUMBER>

Specifies the queue number. Range: 0 to 7.

weight *<WEIGHT>*

Specifies the scheduling weight. Range: 1 to 253.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Assigning WFQ with a weight of **17** to queue **2** in the schedule profile **myschedule**:

```
switch(config)# qos schedule-profile myschedule
switch(config-schedule)# wfq queue 2 weight 17
```

Deleting WFQ for queue **2** from the schedule profile **myschedule**:

```
switch(config)# qos schedule-profile myschedule
switch(config-schedule)# no wfq queue 2
```

Accessing Aruba Support

Aruba Support Services	**https://www.arubanetworks.com/support-services/**
Aruba Support Portal	**https://asp.arubanetworks.com/**
North America telephone	1-800-943-4526 (US & Canada Toll-Free Number) +1-408-754-1200 (Primary - Toll Number) +1-650-385-6582 (Backup - Toll Number - Use only when all other numbers are not working)
International telephone	**https://www.arubanetworks.com/support-services/contact-support/**

Be sure to collect the following information before contacting Support:

- Technical support registration number (if applicable)
- Product name, model or version, and serial number
- Operating system name and version
- Firmware version
- Error messages
- Product-specific reports and logs
- Add-on products or components
- Third-party products or components

Other useful sites

Other websites that can be used to find information:

Airheads social forums and Knowledge Base	**https://community.arubanetworks.com/**
Software licensing	**https://lms.arubanetworks.com/**
End-of-Life information	**https://www.arubanetworks.com/support-services/end-of-life/**
Aruba software and documentation	**https://asp.arubanetworks.com/downloads**

Accessing updates

To download product updates:

Aruba Support Portal

https://asp.arubanetworks.com/downloads

If you are unable to find your product in the Aruba Support Portal, you may need to search My Networking, where older networking products can be found:

My Networking

https://www.hpe.com/networking/support

To view and update your entitlements, and to link your contracts and warranties with your profile, go to the Hewlett Packard Enterprise Support Center **More Information on Access to Support Materials** page:

https://support.hpe.com/portal/site/hpsc/aae/home/

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Regulatory information

To view the regulatory information for your product, view the *Safety and Compliance Information for Server, Storage, Power, Networking, and Rack Products*, available at **https://www.hpe.com/support/Safety-Compliance-EnterpriseProducts**

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